SciOly 2025 Summer Science Study Session
Designer Genes Key **_____ / 58 pts**
By Rasia Yu

Enjoy!
Tiebreaker order: 2, 9, 6, 35, 28c,13, 36

Part A: Miscellaneous MCQ (20 pts), each MCQ is worth 1 point

1. Which of the following organelles does NOT contain DNA?
    a. Mitochondria
    b. Chloroplast
    c. Endoplasmic reticulum
    d. Nucleolus
    e. Nucleus

2. Which of the following is NOT a difference between RNA and DNA?(hint: be VERY specific)(**Tiebreaker #1**)
    a. RNA has Uracil, DNA has Thymine
    b. RNA has to be single stranded, DNA has to be double stranded
    c. RNA has ribose, DNA has deoxyribose
    d. RNA is easier to hydrolyze compared to DNA

3. What is the Southern blot used for?
    a. Analyzes DNA
    b. Analyzes RNA
    c. Analyzes proteins
    d. Analyzes carbohydrates

4. What does CRISPR stand for?
    a. Clustered Regularly Interspaced Short Palindromic Repeats
    b. Clustered Repeatedly Interrupted Short Palindromic Repeats
    c. Clustered Repeatedly Interspaced Slick Palindromic Repeats
    d. Crazy name for some chips company, idk
5. What exactly is CRISPR?
    a. A DNA sequence
    b. A type of small guide RNA
    c. An enzyme
    d. An endonuclease protein

6. What protein acts as the HOMOLOG of FtsZ rings in mitosis?(**Tiebreaker #3**)
    a. Actin(Analog of FtsZ)
    b. Myosin(Analog of FtsZ)
    c. Tubulin(Homolog of FtsZ)
    d. Titin(random muscle protein, not that relevant to the question)

7. Which of the following is a step of PCR?

a. Extension
b. Elongation
c. Termination
d. Expansion

8. If a DNA plasmid is cut using an endonuclease, what will the result be?
    a. Nicked DNA
    b. Circular DNA
    c. Linear DNA, 1 fragment
    d. Linear DNA, 2 fragments

9. Assume I have a restriction enzyme called ILOVESCIOLY, which of the following complimentary fragments is most likely to form sticky ends when spliced?(**Tiebreaker #2**)
    a. 5'-GAATTC-3' and 3'-CTTAAG-5'
    b. 5'-GGCC-3' and 3'-CCGG-5'(Palindromic but no sticky ends)
    c. 5'-GGAC-3' and 3'-CCTG-5'(not palindromic)
    d. 5'-GCCG-3' and 5'-CGGC-3'

10. What does ORF stand for in the context of biology?
    a. Open Reading Frame
    b. Oral ReFlux
    c. Oral Response Framework
    d. Original Reference Factor

11. Which polymerase is used in PCR **when high accuracy is preferred**?
    a. Taq polymerase
    b. Pfu polymerase
    c. DNA polymerase I
    d. DNA polymerase III

12. What Polymerase is used in eukaryotic DNA replication?
    a. DNA pol I
    b. RNA pol II
    c. DNA pol II
    d. DNA pol alpha

13. DNA polymerases read from 3' → 5', RNA polymerase adds bases from 5' → 3', what direction do ribosomes read?(**Tiebreaker #6**)
    a. 3' → 5'
    b. 5' → 3'
    c. Ribosomes don't read, they add bases from 3' → 5'(Ribosomes don't add bases, they use tRNA to grow the peptide chain, so I guess instead of primes, you'll use N and C terminals instead)
    d. Ribosomes don't read, they add bases from 5' → 3'(Ribosomes don't add bases, they use tRNA to grow the peptide chain)

14. In a lac operon, which of the following scenarios will allow the operon to reach MAXIMUM expression?

a. Increasing lactose, CAP is binded on CAP site
b. Decreasing lactose, CAP is not binded on CAP site
c. Increasing lactose, CAP is not binded on CAP site
d. Decreasing lactose, CAP is binded on CAP site

15. The trp operon is classified as a _____ operon:
    a. Repressible
    b. Inducile
    c. Reducible
    d. Inducible

16. Which of the FOLLOWING amino acids only has 1 codon coding for it?
    a. Tyrosine
    b. Cysteine
    c. Phenylalanine
    d. Selenocysteine

17. Oh No. A point mutation has occurred. But miraculously, the protein sequence didn't change at all during translation. What could've happened?
    a. Wobble
    b. Frameshift mutation
    c. Nonsense mutation
    d. Sickle cell Anemia
18. _____ are spliced out of pre-mRNA during _____.
    a. Exons ; alternative splicing
    b. Introns ; alternative splicing
    c. Exons ; translation
    d. Introns ; translation
19. In the Meselson Stahl experiment they discovered that DNA:
    a. Was genetic material
    b. Replicated semiconservatively
    c. Was in a double helix
    d. Has a negative charge
20. Sanger sequencing uses a special type of _____ called "_____", these terminate elongation of the DNA sequence when added to the growing strand:
    a. ATP, GTP
    b. dNTP, ddNTP
    c. GTP, ATP
    d. ddNTP, dNTP

Part B: Punnett squares and MATH (17 pts)

21. At Palo Alto High school, there is a major seagull population. Biology students at PALY discovered an allele that alters the bird's taste preferences called ***s***. Seagulls with homozygous dominant alleles crave savory school lunches while seagulls with homozygous recessive genotypes attack students for sweets. If the seagull is heterozygous for the allele, they enjoy BOTH sweet and savory school lunches, no students are safe.



    a. If 49% of the bird population enjoyed only sweet foods, and assuming the population is in Hardy Weinberg equilibrium, what percentage of the population enjoys ONLY SAVORY FOOD?(Show your work, 3pts)

    p^2+2pq+q^2=1
    p+q=1
    q^2=0.49, q=0.7, p=0.3, p^2=**0.09**
    Correct answer is: 9%

    [If correct answer: +1 pt, If correct answer & work is shown: +3pts, if incorrect answer & work is shown: 0pts, but if they wrote a funny joke: +0.01pt]

    b. What percentage of the seagull population enjoys stealing both savory and sweet food?(Show your work, 2pts)
    1-0.09-0.49=0.42
    Correct answer is: 42%

22. For the past few years, George has been collecting drosophila for his own personal project and breeding them in hopes of becoming the next Mendel. He has red-eyed drosophila and white-eyed drosophila.
    a. Pink-eyed drosophila is a mix of red-eyed and white-eyed drosophila. Would this be an example of codominance or incomplete dominance?(2pt)

    b. In another hypothetical scenario, let's assume that the allele coding for red eyes is sex-linked dominant. A male with red eyes was crossed with a female with white eyes. What percentage of the male offspring would have white eyes?(2pts)

50%

23. George picked up another hobby this past week: bird breeding. He got a blue bird the other day and wants to figure out the phenotype of the bird, so he performs a TEST CROSS. In this particular species, having blue feathers is the dominant trait (Bb or BB) while having white feathers is recessive (bb). Which of the following results of the test cross matches if the bird's phenotype is Bb? (2pts)
    a. 25% blue, 75% white

b. 25%white, 75% blue
c. 50%blue, 50%white
d. 100%blue, 0%white

24. Which of the following results of the test cross matches if the bird's phenotype was BB? (2pt)
    a. 25% blue, 75% white
    b. 25%white, 75% blue
    c. 50%blue, 50%white
    d. 100%blue, 0%white

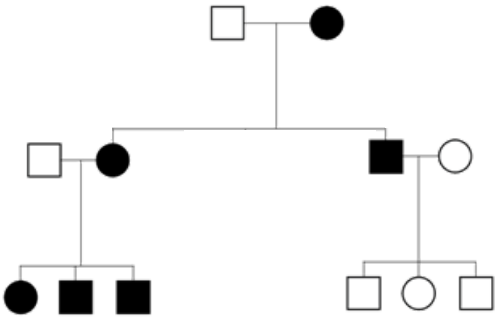
25. The y allele codes for the color of peas, while r codes for whether the pea is wrinkly or not. If two individuals, Yyrr and yyRr were to cross, what percentage of their offspring have the YyRr phenotype?(Show your work & draw a punnett square)(4pts)

	Y	y
y	Yy	yy
y	Yy	yy

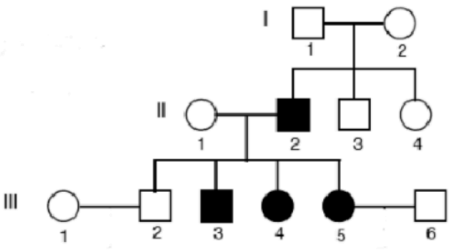
+2pts for punnett square

= ½,
½ *½ =¼ Convert to percentage = **25%**
+1pts for work shown, +1pts for correct answer

Part C: Playing with pedigrees and pictures(21 pts)

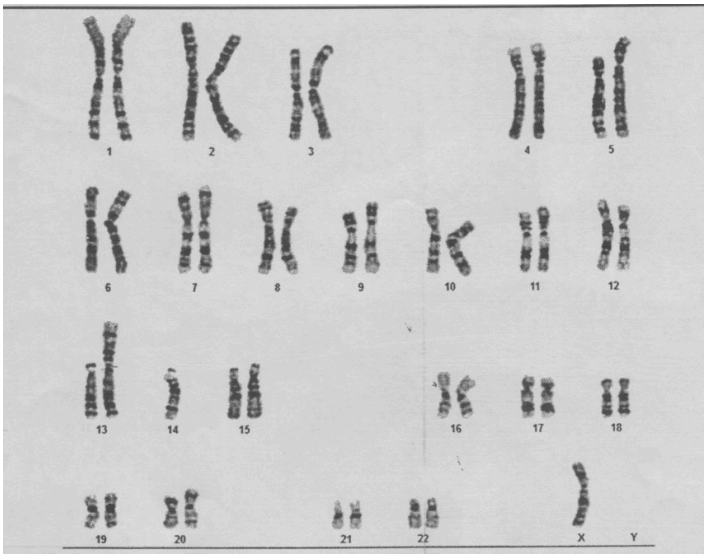


26. What type of inheritance is this?(on the right, 2 pts)
    a. Autosomal recessive
    b. Autosomal dominant
    c. X-linked recessive
    d. X-linked dominant
    e. Mitochondrial
    f. Y-linked



27. What type of inheritance is this?(on the left, 2 pts)
    a. Autosomal recessive(PFU. You can tell because it skips a generation)
    b. Autosomal dominant
    c. X-linked recessive
    d. X-linked dominant
    e. Mitochondrial
    f. Y-linked

28. What is the name of the image below called? _____(karyotype 1pt)

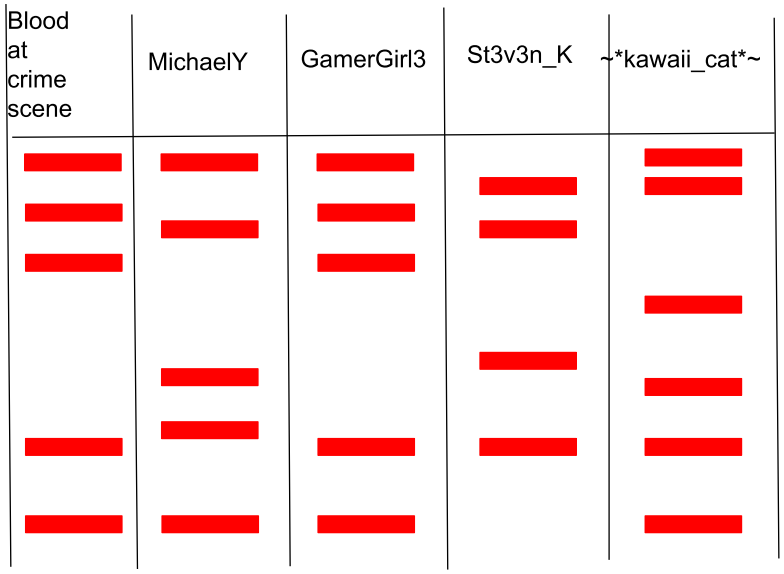


a. When during mitosis is this image typically taken? _____(Metaphase, 1pt)
b. What genetic disorder does this individual have?_____(Turner's syndrome, 2pts)
c. What is the sex of this individual?_____(female, 2pts)(**Tiebreaker #5**)

29. What does RFLP stand for?(1 pt)

Restriction fragment length polymorphism

30. Whose blood was at the CRIME????????(1pt)



It was GamerGirl3 :p

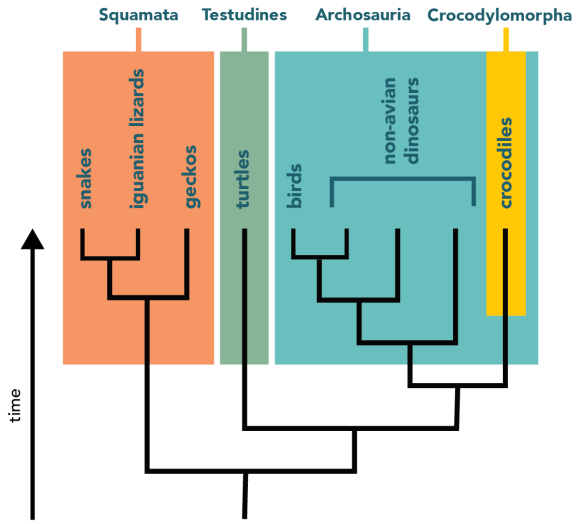
31. Translate this sequence of mRNA please(codon table is provided, 5pts):
5' AGUAUGUCCCCUAUCCAAAGAAAACGGUAGGGGUCG 3'
AGU AUG UCC CCU AUC CAA AGA AAA CGG UAG GGG UCG
Met - Ser - Pro - Ile - Gln - Arg - Lys - Arg -Stop

First letter	Seond letter: U	Seond letter: C	Seond letter: A	Seond letter: G	Third letter
U	UUU Phe UUC Phe UUA Leu UUG Leu	UCU Ser UCC Ser UCA Ser UCG Ser	UAU Tyr UAC Tyr UAA Stop UAG Stop	UGU Cys UGC Cys UGA Stop UGG Trp	U C A G
C	CUU Leu CUC Leu CUA Leu CUG Leu	CCU Pro CCC Pro CCA Pro CCG Pro	CAU His CAC His CAA Gin CAG Gin	CGU Arg CGC Arg CGA Arg CGG Arg	U C A G
A	AUU Ile AUC Ile AUA Ile AUG Met	ACU Thr ACC Thr ACA Thr ACG Thr	AAU Asn AAC Asn AAA Lys AAG Lys	AGU Ser AGC Ser AGA Arg AGG Arg	U C A G
G	GUU Val GUC Val GUA Val GUG Val	GCU Ala GCC Ala GCA Ala GCG Ala	GAU Asp GAC Asp GAA Glu GAG Glu	GGU Gly GGC Gly GGA Gly GGG Gly	U C A G

32. The analog of the TATA box in prokaryotes is the_____(pribnow box, 2pts)

33. Sickle cell disease is a result of a point mutation in a 5'-GAA-3' codon. What amino acid does glutamic acid get turned into in individuals with Sickle cell anemia?_____ (Valine, 1pts)

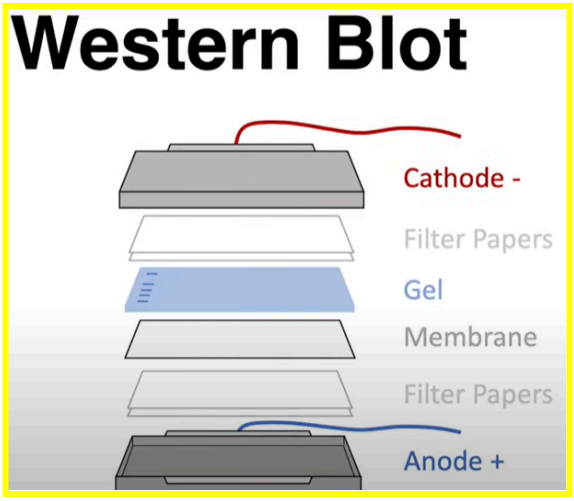
34. The image below is of a phylogenetic tree. Select all of the follow groups which are monophyletic(2pts):
    a. Squamata +0.5pts
    b. Testudines +0.5pts
    c. Archosauria (if selected, -0.5pts)
    d. Crocodylomorpha +0.5pts



35. What is the purpose of using SDS in Western blotting?(3pts)**(Tiebreaker #4)**

Mainly to make all of the protein the same charge when on the polyacrylamide gel(because the charges from the amino acid side chains could affect results otherwise). The goal of using the SDS page is to find the protein's molecular weight.

36. Is the cathode in Western blotting positively or negatively charged?(2pts)**(Tiebreaker #7)**



CONGRATULATIONS!!! YOU ARE DONE!!!!

Extra credit (1pt)
Write your own Designer gene questions based off of what you learned in the past week:
If the question actually follows Designer gene rules and is informationally accurate(according to google), +1 pt.
But if it's only funny, +0.1pt.